Pannala Aruna Paint Center — POS System Plan

Oct 25, 2026 · @KING_UPE

Summary

Pannala Aruna Paint Center gets a keyboard-first web POS that runs on a shop server and mirrors to the cloud for admins.

- **Three sections:** Stock (suppliers, purchases, products, mix paint, reorder), Billing (invoices, quotations, returns, day end) and Admin (dashboard, reports, commission, monthly closing, users, settings).
- **Four roles, combined per user:** the Super Admin ticks Admin, Stock Manager or Cashier for each user, so one person can hold several.
- **Sub-products:** every item has a main code and sub-products (10001-01, 10001-02…), one per supplier, cost, price and expiry date. The system creates and reuses them automatically.
- **Mix paint:** cans become liters of mix paint (MP-code), sold loose by the liter.
- **Cash only:** no credit, cards or cheques at the counter. Items are found by typing a code or a name.
- **Private numbers:** totals of cost, profit and margin are visible only to the Super Admin; each product's cost, profit and margin are also visible to Stock Managers.
- **Nothing is lost:** every bill, reprint, return, price change and stock move is logged, and records are never deleted.

Your answers and the 12 requirement updates from 25 Oct 2026 are built into every section. Every question is now answered.

Users, roles and permissions

The Super Admin registers every user and ticks the roles each one gets; a user can do everything their ticked roles allow.

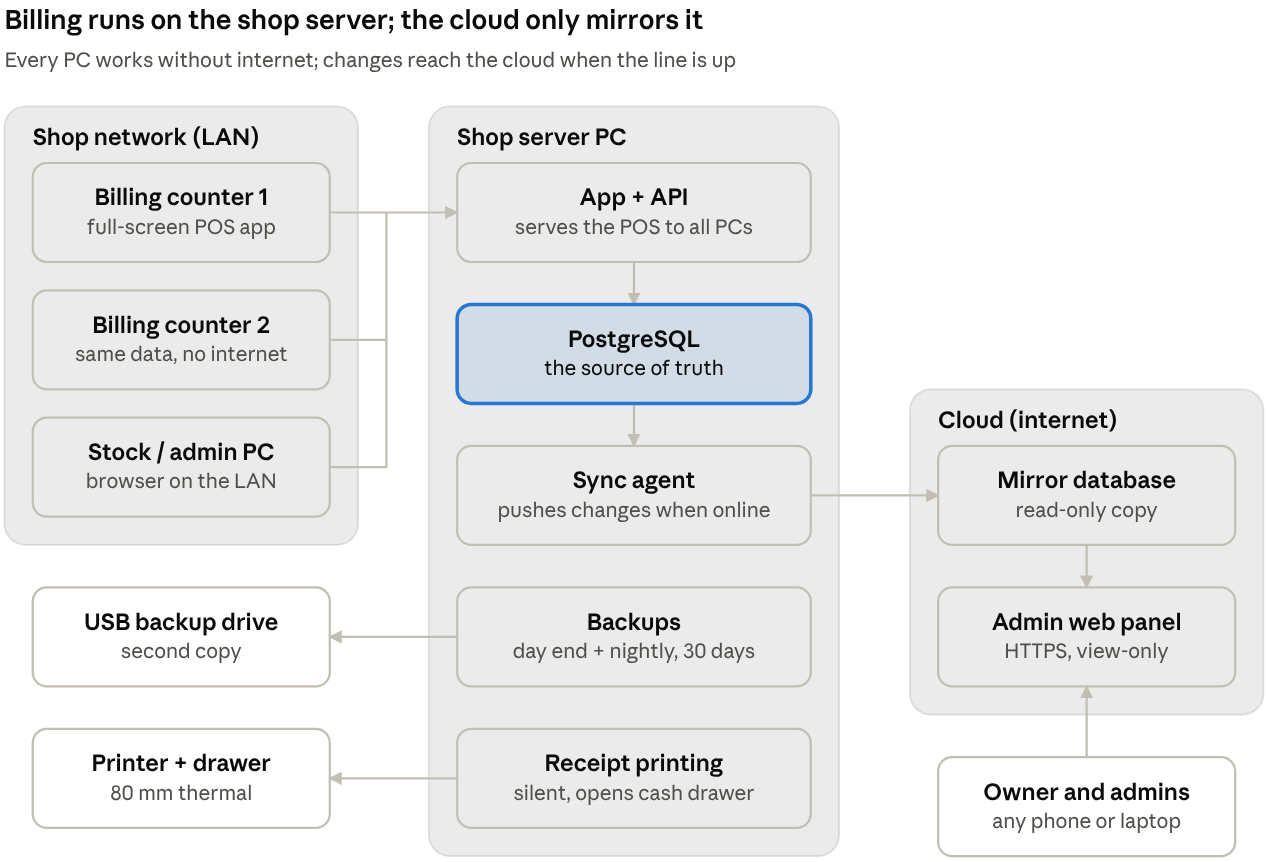
Role	Can use	Needs approval for
Super Admin	Everything, including total cost, total profit and margin, monthly closing, users, roles, settings, backups and restore	—
Admin	Dashboard and reports without cost or profit, commission, stock, billing, value manager, approvals	—
Stock Manager	Suppliers, purchases, products with each item's cost, profit and margin, mix-paint transfers, reorder, stock and supplier reports	Write-offs (Admin)
Cashier	Billing, quotations, bill history, reprints, returns, loose sales from mix paint, own counter's day end	Changing a mix paint price or voiding a bill (Admin PIN); opening a can into mix paint (Stock Manager or Admin PIN)

- **Roles by checkbox:** at registration the Super Admin ticks one or more roles, for example Stock Manager and Cashier together. Changing them later works the same way.
- **New roles:** the Super Admin can also build a role from module permissions (view, create, edit, approve, export per screen). It then appears as another checkbox. A role still ticked for any user can't be deleted.
- **Safety rules:** at least one active Super Admin must always exist, and only a Super Admin can tick the Super Admin role.
- **Manager override:** when a cashier hits a limit, an Admin types their PIN on the same screen. Both names go into the log.
- **Login:** username and password only. Cashiers switch user with a 4–6 digit PIN, and the screen locks after a set idle time.
- **Passwords:** the Super Admin resets passwords and PINs; the user must change a reset password at next login.
- **Recovery:** printed recovery codes at setup, so a forgotten Super Admin password never locks the shop out.

- **Cost privacy:** totals of cost, profit and margin (shop-wide, by category or by period) appear only for the Super Admin. Stock Managers also see each product's cost, profit and margin. Admins and Cashiers see neither.

Architecture, sync, install and backups

The shop server is the single source of truth: billing never waits for the internet, and the cloud is a read-only mirror for admins.



system architecture · counters, shop server, cloud mirror

Counters and the stock PC talk only to the shop server; the sync agent copies every change to the cloud mirror that admins read.

How it runs

- **Shop server:** one Windows PC runs the app, the PostgreSQL database, receipt printing, backups and the sync agent. It can be the main counter PC or a small dedicated PC.

- **Counters:** 1–2 billing PCs open the POS over the shop network in a small full-screen desktop shell (Tauri or Electron) that loads the server's page. Every key reaches the app, and it starts with Windows.
- **Locations:** the system starts with one shop. The Super Admin can add shops or warehouses later; every record already carries its location, so nothing needs migrating.
- **Cloud:** a mirror database plus the view-only admin web panel at a private web address.

Sync

- Every change writes a row to a sync outbox inside the same database transaction, so nothing can be missed.
- When the internet is up, the sync agent pushes changes at once and checks again every 15 seconds. The cloud applies each change only once, so retries never double anything.
- The status bar shows "Online · synced 10 s ago" or "Offline · 128 changes waiting".
- Online edits, if wanted later, are stored in the cloud as commands that the shop server pulls and applies. The shop stays the only writer, so there are no sync conflicts.

Backups

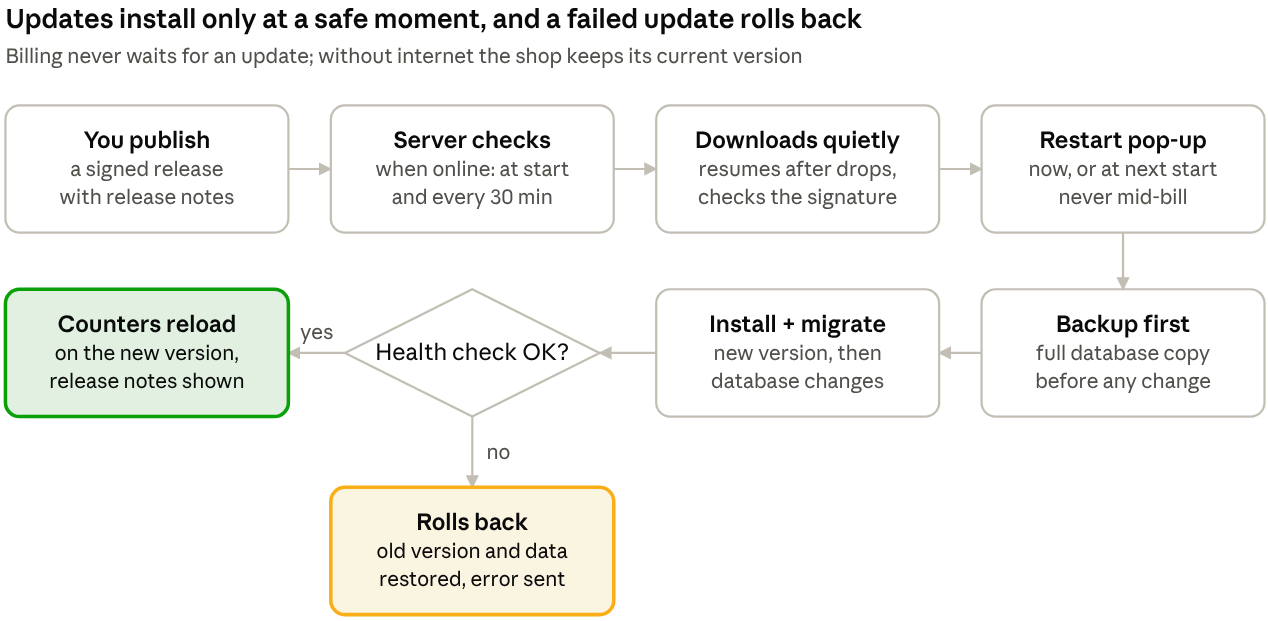
- Automatic backup at day end and every night at a set time, compressed and named by date.
- Keeps 30 daily and 12 monthly copies, a second copy on a USB or second drive, and a copy in the cloud when online. Backups are not encrypted.
- Super Admin restore wizard, one test restore a month, and backup status on the dashboard.

Install and updates

- One Windows installer sets up the server, database and services, which start when the PC boots.
- First-run wizard: shop details and logo, Super Admin account and recovery codes, the first shop, printer test, backup folder, cloud link.
- Updates install themselves; see Automatic updates below.

Automatic updates

When you release a new version, the shop installs it by itself once it is online and at a safe moment; nobody at the shop has to do anything.



automatic update flow · 8 steps, 1 check

Bills never wait on an update: it installs on a restart from the pop-up or at the next start, and a failed check puts the shop back exactly as it was.

- **Release feed:** you upload each build to the cloud server with its version number, release notes and signature. Shops read that feed.
- **When it checks:** at start, every 30 minutes while online, and right after the internet comes back.
- **Download:** in the background and resumable on patchy internet, so billing never slows down or waits.
- **Install time:** once downloaded, a pop-up asks to restart the application now to update. If it is cancelled, the update installs at the next close and open. The pop-up never interrupts an open bill.
- **One update for all PCs:** only the shop server is updated. Counters reload the new screens by themselves, and the desktop shell updates itself the same way when it changes.
- **Safety:** the server installs only releases signed with your key. A full backup runs first; if the health check fails, the old version and data come back and the error is sent to you.

- **Database changes:** they run automatically during the install, after being tested on a copy of real shop data before release.
- **Older shops keep syncing:** the cloud's sync connection is versioned, so a shop that hasn't updated yet still syncs.
- **Version shown:** in the bottom bar and the admin panel, with an update history (version, date, result) and the release notes after each update.

Hardware

- An 80 mm thermal printer with a cash drawer at each counter. Receipts and quotations print straight to it, with no print dialog.
- Any bill can also be exported as an A4 PDF; reports and reorder lists are PDFs too.
- A UPS on the server PC, router and printer keeps billing running through short power cuts.

Security

- The shop server is reachable only on the shop network.
- The cloud panel uses HTTPS with username and password login, without two-factor login or request rate limits.
- With those removed, passwords carry all the protection: at least 8 characters, and one account per person.
- The server checks permissions on every request, not just in the screens.
- Clock guard: when online, the PC time is compared with the cloud. A gap over 2 minutes raises a warning, because bill dates depend on it.

Tech stack and data rules

One TypeScript codebase runs at the shop and in the cloud; a setting switches it between full mode and view-only mode.

Layer	Choice	Why
Screens	React + Vite, Tailwind CSS, shadcn/ui	Fast app served by the shop server; clean rounded components
Lists and forms	TanStack Query and Table, React Hook Form + Zod	Quick grids and validated forms

Layer	Choice	Why
Keyboard	Own hotkey layer with per-user key maps, cmdk command palette	Zero-mouse operation; each user can remap keys
Charts	Recharts or ECharts	Dashboard and report charts
Server	Node.js + Fastify, Prisma	Typed API; WebSocket pushes live stock changes to every counter
Database	PostgreSQL at the shop and in the cloud	Same schema at both ends, crash-safe, strong constraints
Validation	Shared Zod schemas	The same rules in the screen and on the server
PDF and Excel	pdfmake or React-PDF, ExcelJS	Reports without bundling a browser
Receipts	ESC/POS printing from the server	Silent printing and cash drawer; Sinhala printed as an image
Install	Inno Setup or NSIS installer; Node and PostgreSQL as Windows services	One-click setup
Cloud	Small VPS or managed host with PostgreSQL	Mirror database and admin panel

Next.js isn't needed here. The POS is an app behind a login, so a Vite app plus an API is lighter to install on a shop PC.

Data rules

- IDs are UUIDv7 (time-ordered and safe to sync). Readable document numbers come from sequences on the shop server, in formats the Super Admin sets with letters allowed, such as INV-004512 or A26-INV-0125.
- Money is Decimal(14,2) and quantities Decimal(14,3), never floating point.
- Stock is a ledger: every in or out is a stock-movement row with its cost and document. Current stock is a cached balance updated in the same transaction and checked nightly.
- Each bill line stores the name, prices and cost at the moment of sale, so old bills and profit reports never change.
- Anything with history is deactivated, never deleted.

- Every stock change runs in one database transaction with row locks, so two counters can't sell the last can twice.

Value manager

One small screen manages every dropdown list, so the same value is never typed two different ways.

List	Used in	Examples
Categories	Products, reports	Emulsion, Enamel, Thinner, Putty, Brushes
Brands	Products, reports	Your paint and hardware brands
Units	Products	piece, box, can, L, kg
Locations	Purchases, stock, billing	Shop, plus a built-in Damaged location; more shops or warehouses later
Sales reps	Purchases (only the chosen supplier's reps)	Name and phone, linked to a supplier
Reasons	Returns, adjustments, voids, cash out	Damaged, Expired, Wrong item, Count correction, MP wastage, Petty expense
Cutoff types	Monthly closing	Salary, Rent, Bills, Savings

- Names are unique, ignoring capital letters and extra spaces.
- A value in use can't be deleted, only deactivated. Two duplicates can be merged into one, history included.
- Typing a new value in any dropdown offers "+ Add" right there, if the role allows it.
- Only the Super Admin adds shops or warehouses.
- Each list has its own order and an active switch.

Products and sub-products

Every item has one main product and one or more sub-products, one per supplier, cost, price and expiry date. Purchases create and reuse them automatically, so nobody types sub codes by hand.

Code	What it is	Supplier	Unit	R-price	Sales price
10001	Thinner 4L, main product	—	can	—	—
10001-01	Sub-product	Supplier A	can	Rs 2,400	Rs 2,300
10001-02	Sub-product, new price	Supplier A	can	Rs 2,600	Rs 2,450
10001-03	Sub-product	Supplier B	can	Rs 2,500	Rs 2,350
10001-04	Same prices as -02, new cost	Supplier A	can	Rs 2,600	Rs 2,450

A product sold loose also gets a mix paint item counted in liters: White Emulsion 4L (20001) feeds MP-20001. Several sizes of one paint (1 L, 4 L, 10 L) can be linked to one mix paint item, each converting by its own pack size.

Main product fields: code (old code on import, otherwise 5 digits; letters allowed), name, short print name for the receipt, description, category, brand, unit, pack size (4 L per can, for liter conversion), buying unit and pieces per box (for items bought in boxes and sold singly), reorder level and reorder qty, allow loose sale, mix paint item, track expiry, active.

Sub-product fields: code (main-01, main-02…), name and description (copied from the main product, editable), supplier, market value, purchase discount, cost, actual price, R-price, W-price, sales discount, sales price, expiry date, first received date, stock per location.

Automatic rules

1. A posted purchase line looks for a sub-product of that main product with the same supplier, cost, R-price, W-price, sales price and expiry date.
2. If one matches, the stock is added there. If anything differs, including a new cost or a new expiry date, the next code is created (10001-05).
3. A matching sub-product at zero stock is reused instead of making a new code.
4. A sub-product with zero stock in every location disappears from billing and search. It is never deleted, and a return or a purchase brings it back.
5. Changing a main code (10001 to 10050) renames its sub-product and mix paint codes after a confirmation; history stays linked.
6. Items bought in boxes are stocked and sold as single pieces: 2 boxes of 12 enter stock as 24 pieces, each costing the box cost ÷ 12.
7. Admins can change the selling prices of one sub-product, or of all sub-products of a main product at once. Every change is kept in price history.

Number formats

The Super Admin can change every format, letters included; these are the starting formats.

Record	Format	Example
Supplier	SUP-NNNN	SUP-0042
Main product	5 digits (old codes kept)	10001
Sub-product	main-NN	10001-02
Mix paint	MP-main	MP-20001
Purchase (GRN)	GRN-YYYY-NNNNN	GRN-2026-00125
Invoice	INV-NNNNNN	INV-004512
Quotation	QT-NNNNNN	QT-000318
Sales return	SR-NNNNNN	SR-000077
Purchase return	PR-NNNNNN	PR-000012
Supplier payment	SP-NNNNNN	SP-000230
Transfer	TR-NNNNNN	TR-000051
Mix-paint transfer	MPT-NNNNNN	MPT-000019
Adjustment	ADJ-NNNNNN	ADJ-000008
Day end	DE-YYYYMMDD-counter	DE-20261024-C1

Invoice and other document numbers can mix letters and digits, and every search box accepts both.

Stock management

The Stock section takes goods from supplier to shelf, and every action writes to the stock log and the supplier log.

Suppliers

Fields: code (auto), name, 1–3 phone numbers, address, description, email, credit days, opening balance, sales reps (name and phone), active.

The balance is calculated, never typed. A positive balance means the shop owes the supplier.

```
\text{Balance} = \text{Opening} + \text{Purchases} - \text{Payments} -
\text{Purchase returns} \pm \text{Adjustments}
```

A supplier with history can be deactivated but not deleted.

Supplier activity log

- Filters: supplier, start date, end date and entry type (purchase, payment, return, adjustment).
- Shows the opening balance on the start date, then each entry with date, type, document no, debit, credit and running balance, then the closing balance.
- Enter on a row opens the document. Export to PDF or Excel, or print.

Purchases (GRN)

Header: GRN no (auto), supplier invoice or receipt no, date, ref (a free-text note), supplier, location, sales rep (only that supplier's reps).

Lines

Field	Type	Rule
Item code / name	entered, with suggestions	Main product; the system picks or creates the sub-product. A new item can be created inline.
Value (market price)	entered	Price per unit on the supplier's list (per box for boxed items)
Qty	entered	More than 0; boxes convert to pieces
Free	entered	0 or more
Discount	entered	% or Rs per unit; this is the purchase discount
Cost	calculated	Value − Discount
Amount	calculated	Cost × Qty
Actual price	calculated	Amount ÷ (Qty + Free)

Field	Type	Rule
R-price	calculated, editable	Equals Value by default
W-price	entered	Between Actual price and R-price
Sales discount	entered	% or Rs; not above the purchase discount. It becomes the default discount at billing.
Sales price	calculated, editable	R-price − Sales discount; never below Cost
Margin	calculated	Sales price − Actual price, in Rs per unit, so free items count
Exp date	entered	After the purchase date; a new date makes a new sub-product
Category, Brand	from the main product	Set when the main product is created

Worked example: Value Rs 2,000, Qty 10, Free 2 and Discount 10% give Cost Rs 1,800, Amount Rs 18,000 and Actual price Rs 1,500 (18,000 ÷ 12). With R-price Rs 2,000 and a 5% sales discount, the Sales price is Rs 1,900 and the Margin Rs 400 per unit (1,900 − 1,500, free items included). At billing the cashier can give up to Rs 200 off, the purchase discount.

Footer and posting

- Subtotal, invoice discount, other charges (transport) and net total.
- Invoice discount and transport stay on the invoice: they change the net total and the supplier balance, never the item costs.
- The supplier's invoice total is typed in and must match the net total within Rs 1 before posting. This catches typing mistakes.
- Paid now (any amount); the rest goes to the supplier balance.
- Save as draft (no stock change) or post (stock and supplier balance update).
- The same supplier with the same invoice no raises a warning.
- A posted purchase can be edited only while none of its stock has moved. Otherwise an Admin reverses it and re-enters it.
- Keyboard: Enter moves across the line, Enter on the last column starts a new line, and Ctrl+Enter posts.

Purchase returns

- Choose the supplier (and the GRN if known), the sub-products, the qty up to the stock available, and a reason.
- Stock and the supplier balance go down, and a return note prints.

Supplier payments

- Cash, cheque or bank transfer. A cheque records its number, bank and date.
- Post-dated cheques go into a cheque register with reminders before their date.
- A payment reduces the balance, oldest bills first. Paying more than the balance shows a warning and records an advance.

Stock view and editing

- Main products with total stock; Enter opens their sub-products and the stock in each location.
- Filters: category, brand, supplier, location, low stock, out of stock, expiring soon, and hidden zero-stock sub-products.
- Edit name, code and description on main products and sub-products; every edit is logged.
- Item card: every movement of an item with a running balance (bin card).
- Each product's cost, profit and margin show for Stock Managers and the Super Admin; totals only for the Super Admin.

Location transfers

- Available once the Super Admin adds a second shop or warehouse.
- From location, to location, sub-products and qty (up to what the source holds), and a note. It posts at once and prints a transfer note.
- Billing then offers "Move 1 from Store?" when the Shop has none and the role allows it.

Transfer to mix paint

- Stock Managers and Admins only. Choose the can's sub-product and the number of whole cans; the system converts them to liters (1 can × 4 L = 4 L).
- The liters go into the product's mix paint item, MP-<main code>, or into the one its size is linked to.

- The screen shows the mix paint liters and its price per liter in a box; the average cost per liter shows for Stock Managers and the Super Admin.
- Keep the price or change it, but never below the new average cost.
- New average cost per liter = (old liters × old cost + new liters × new cost) ÷ total liters.

Example: MP-20001 holds 2 L at Rs 500 per L. Adding one 4 L can with an actual price of Rs 1,800 (Rs 450 per L) makes the new average (2 × 500 + 4 × 450) ÷ 6 = Rs 466.67 per L.

Dried or spilled mix paint is recorded as an adjustment with the reason "MP wastage".

Adjustments and stock count

- Adjustment: add or remove qty of a sub-product, or liters of mix paint, with a reason. Write-offs need Admin approval.
- Stock count: print a count sheet by location or category, enter counted qty, review the difference in qty and rupees, then an Admin posts it.
- Stock can never go below zero.

Reorder

- Each main product has a reorder level and reorder qty, which the system can suggest from sales speed.
- Columns: product, stock (all locations), reorder level, average daily sales over 30 days, days of stock left, on order, suggested qty, last supplier, order qty. Last cost shows for Stock Managers and the Super Admin.
- For an item below its level, suggested qty = average daily sales × (lead days + cover days) − stock − on order. It is rounded up to whole packs and is at least the reorder qty.
- Filters: supplier, brand, category, location, below level, out of stock, runs out within N days.
- Space selects rows. "Create reorder list" makes one PDF per supplier with items and qty, ready to print or share.
- The list is marked Ordered so the next suggestion subtracts it. When the goods arrive, the purchase screen can pre-fill from it.
- Shelf refill, once there are more locations: items low in the Shop but available elsewhere, moved in one step.

Stock reports

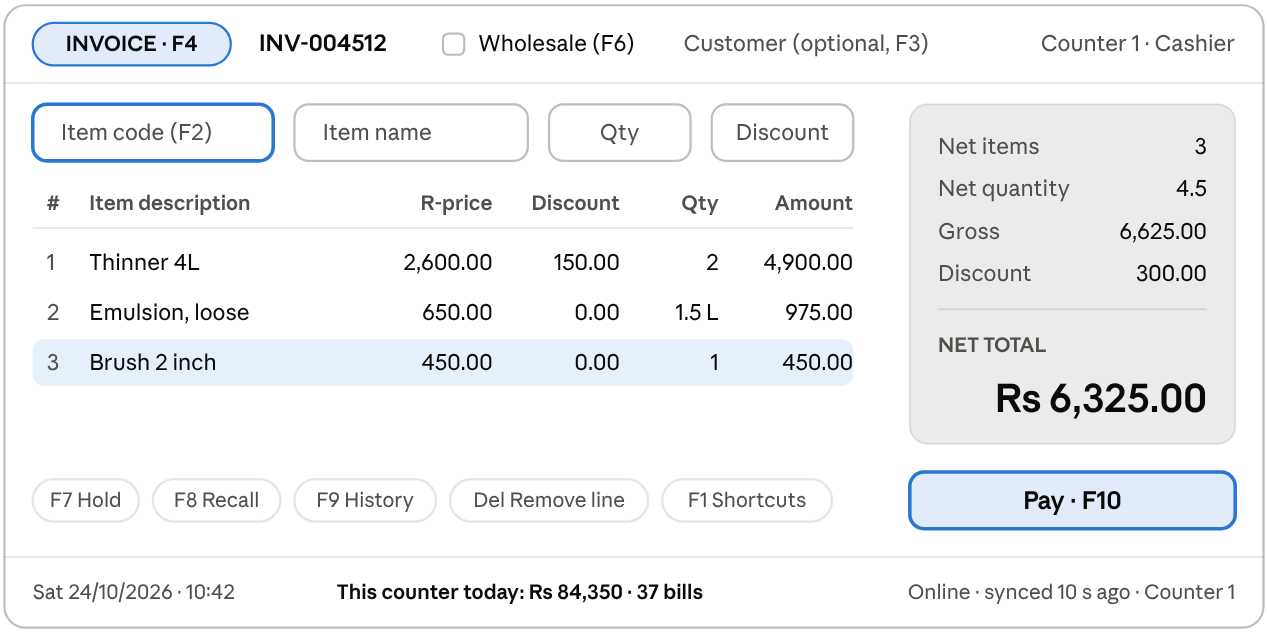
Current stock by location and sub-product, stock value at sales price (at cost for the Super Admin), item movement, low stock, near expiry and expired, dead and slow stock, transfers, adjustments, mix paint movements and stock-count differences. Every report filters by date and exports to PDF or Excel.

Billing

The billing screen is built for one hand on the keyboard: type, Enter, repeat, then F10 to pay.

Billing screen: totals and keys always in view

Example bill: two cans of thinner, 1.5 L of loose emulsion and one brush



billing screen mock-up · example bill

The newest line is highlighted, and the net total and Pay key sit in the same place on every bill. The keys shown are defaults; each user can remap them.

Screen

- **Top bar:** invoice or quotation (F4), bill no, a small Wholesale checkbox (F6), an optional customer name for quotations (F3), counter and cashier.

- **Entry row:** an Item code field and an Item name field. Typing a code or a name in either opens the same suggestion list; picking an item fills the active row with its code, description, price and default discount.
- **Lines:** #, item description, R-price, discount, qty, amount.
- **Totals:** net items, net quantity, gross, discount and a large net total.
- **Bottom bar:** date, live time, this counter's sales today (for example "This counter today: Rs 84,350 · 37 bills"), sync status, counter.

A sale, start to finish

1. Type a code or a name in the Item code or Item name field, pick with ↑↓ and press Enter.
2. If the item's batches have different prices, the multi-price pop-up opens with the oldest highlighted; press Enter.
3. Type the qty and press Enter. A decimal or an "L" starts a loose sale.
4. The discount is pre-filled from the batch. Enter adds the line and returns to the Item code field.
5. F10 opens payment. Type the cash given, and Enter shows the change.
6. "Print receipt?" appears with Yes selected. Enter prints; choosing No skips printing. Either way the drawer opens and a new bill starts.

Multi-price pop-up

- Opens automatically when the item's in-stock sub-products have two or more different prices.
- Lists every batch: sub code, price, qty in this location, expiry. The oldest stock is highlighted and selected, so Enter sells first-in, first-out.
- No pop-up when only one batch is in stock, or when all batches share one price; the oldest is then used automatically.
- Typing a full sub code (10001-02) skips the pop-up.
- Asking for more than a batch holds fills the rest from the next batch, each line at its own price, such as "2 from 10001-01 at Rs 2,300 and 3 from 10001-02 at Rs 2,450".

Discounts and price checks

- The sales discount set at purchase is filled in by default.
- Type 50 for rupees or 5% for percent.

- A line can get at most its batch's purchase discount, and no line sells below cost.
- Discounts are per line; there is no bill-level discount.
- A price can never be above the R-price.
- With Wholesale ticked, every line takes its W-price with no extra discount, and the receipt shows "Wholesale".

Loose paint

Loose sales sell liters from the product's mix paint stock, and a new can is opened only when it runs short.

1. Choose the paint, for example 20001 White Emulsion 4L, and type "1.5L", or part of a can such as 0.375 (shown as "= 1.5 L").
2. If MP-20001 holds at least 1.5 L, the line sells straight from mix paint at its price per liter.
3. If not, a can must go into mix paint first. Only Stock Managers and Admins move cans, so one of them types their PIN, and the can's 4 L enter MP-20001.
4. The mix paint price per liter is shown. Enter keeps it; changing it up or down needs an Admin PIN.
5. The line reads "White Emulsion, loose · 1.5 L × Rs 650 = Rs 975", and the rest stays in MP-20001.

Why the "L": typing 1 alone sells one can, while 1L sells one liter, so a cashier can't sell the wrong amount by mistake. Mix recipes are not saved; the system tracks only the mix paint stock.

Payment

- Cash only: no cards, cheques, credit or split payments.
- Type the amount given and the change shows in large type. Enter with an empty amount means exact cash.
- The cash given must cover the net total.
- Then "Print receipt?" asks Yes or No, with Yes already selected.

Hold and recall

- F7 parks the current bill, for example while a customer checks colors. F8 lists held bills to resume.
- Held bills don't reserve stock; stock is checked again at payment.

Quotations

- F4 switches to quotation mode on the same screen: QT number, no stock change, and a validity date (default 7 days).
- Quotations print on the 80 mm printer in the same layout as receipts, headed QUOTATION with the validity date. There is no A4 quotation.
- From bill history, C converts a quotation to an invoice. Stock and prices are re-checked, and quoted prices hold until the validity date.

Printing and reprints

- 80 mm receipt: shop name, address and phone, bill no, date and time, cashier, customer name if entered, the lines (description, R-price, discount, qty, amount), net items, net qty, gross, discount, net total, cash given, change, "You saved Rs X", and a footer with the return policy.
- Receipts can also print in Sinhala, sent to the printer as an image.
- Any bill can be exported as an A4 PDF.
- Every later copy prints "REPRINT #2 · date and time · user" and is logged with who, when and which bill.

Bill history

- Search by invoice or quotation no (letters or digits, any part), date range, customer name, cashier or amount.
- Keys on a selected bill: P reprint, R return, C convert a quotation, V void.
- Void: same day only, with a reason and an Admin PIN. The bill stays visible stamped VOID, and its stock comes back.

Returns

1. Press F9, find the invoice and press R.
2. Each line shows the sold qty, the qty already returned and the returnable qty.
3. Select lines with Space, change the qty if needed (up to returnable), and choose a reason and Restock or Damaged.
4. The refund is the cash the customer actually paid for those units, after line discounts.
5. Restocked units go back to the same sub-product they were sold from, and a hidden zero-stock sub-product reappears. Damaged units go to the Damaged location.
6. A return receipt prints and the return is logged.

Returns have no time limit, and Cashiers, Admins and the Super Admin process them without approval. Returns of old-system bills, without a system invoice, are Admin only. Loose and mixed paint can't be returned.

Day end

- Start of day: the opening cash float.
- Cash in and cash out during the day with a reason, such as small daily expenses. Each shows in the day end and the monthly closing.
- Day end shows the expected cash. The cashier counts notes and coins, and any difference is recorded.
- The day-end report prints, then a backup and a sync run.

Admin panel

Admins see the shop's health on one dashboard and can pull any report for one date, a date range or a custom time range. Anything showing cost or profit is for the Super Admin only.

Dashboard

Cards: today's sales, bills today, average bill, expected cash in drawers, returns today, low-stock items, items expiring soon, supplier dues, last sync, last backup and app version. The Super Admin also sees gross profit for today and this month.

Chart	Kind	Answers	Seen by
Sales trend vs previous period	Line	Are sales growing?	Admin, Super Admin
Sales and gross profit by month	Bars + line	Is profit keeping up with sales?	Super Admin
Best sellers by qty and revenue	Horizontal bars	What sells?	Admin, Super Admin
Best sellers by profit	Horizontal bars	What earns?	Super Admin
Sales by category and brand	Bars	Which product lines matter?	Admin, Super Admin

Chart	Kind	Answers	Seen by
Wholesale vs retail sales	Bars	How much sells at W-price?	Admin, Super Admin
Busy hours (hour × weekday)	Heatmap	When to staff and restock?	Admin, Super Admin
Cashier performance	Bars + table	Bills, sales, discounts given, returns	Admin, Super Admin
Stock value by category	Bars	Where is money tied up? At sales price; at cost for the Super Admin	Admin, Super Admin
Supplier dues by age (0–30, 31–60, 61–90, 90+ days)	Stacked bar	Whom to pay first?	Admin, Super Admin
Loose paint liters sold	Line	How much sells loose?	Admin, Super Admin
Loose paint margin	Line	Is loose selling worth it?	Super Admin
Returns trend	Line	Is there a quality problem?	Admin, Super Admin
Same product, different suppliers	Table	Who is cheapest for each item?	Super Admin

Reports

Every report has three date modes: one date, a date range, or a custom date and time range (for example 24/10 08:00 to 24/10 13:00). Presets cover today, yesterday, this week, this month and last month. Every report prints and exports to PDF and Excel, with the shop header, the filters used, who generated it and page numbers.

Group	Reports	Seen by
Sales	Billing report (bill by bill), item-wise, category and brand, cashier-wise, wholesale bills, discounts given, returns, voids and reprints, quotations (and how many converted), loose paint sales, day-end reports	Admin, Super Admin
Commission	Commission by person and period	Admin, Super Admin
Profit	Gross profit by bill, item, category, brand and period; monthly closing	Super Admin
Stock	Current stock, stock value, item movement, low stock, near expiry, dead and slow stock, transfers, adjustments, mix paint movements, stock-count differences	Admin, Super Admin; item costs also for Stock Managers, cost totals Super Admin only
Purchases and suppliers	Supplier activity log, supplier balances by age, purchases by supplier and item, purchase returns, payments	Admin, Super Admin; item costs also for Stock Managers
Control	Audit log, user logins, price change history	Admin, Super Admin

Commission

Admins view and calculate commission without seeing cost or profit. Each staff member earns a rate on their net sales, set by the Super Admin, with a monthly commission report.

Monthly closing and cutoffs

On the month start date the Super Admin sets (the 1st by default, or any day such as the 25th), the system generates the previous month's closing report, applying the Super Admin's fixed and percentage cutoffs in order. Because it shows cost and profit, the closing is for the Super Admin only.

- **Rules (Super Admin):** name, type (fixed Rs or percentage), value, base for a percentage, order, active. The base is what remains after fixed cutoffs, unless a rule picks net sales or gross profit.
- **Variable bills** such as electricity and water start with last month's figures; the Super Admin enters the actual amounts.
- **Daily expenses** taken from the cash drawer flow in from the day ends.

Example with a percentage based on what remains after fixed cutoffs:

Line	Rs
Net sales	2,450,000
Cost of goods sold	−1,960,000
Gross profit	490,000
Daily expenses from day ends	−18,000
Salaries (fixed)	−120,000
Rent (fixed)	−45,000
Electricity and water (actual)	−14,000
Remaining after fixed cutoffs	293,000
Savings, 20% of remaining	−58,600
Shop development, 5% of remaining	−14,650
Balance	219,750

After review, the Super Admin closes the month: the PDF report is saved and the month is locked. Back-dated changes then need a Super Admin unlock, which is logged.

Alerts

Low stock, near expiry (60 days), supplier payments and cheques due, mix paint price changes, high returns, cash differences at day end, failed backup or update, sync waiting too long, day end not done, and a wrong PC clock.

Audit log

Every create, edit, delete, price change, override, reprint, void, return and login is recorded with who, when, which record, and the values before and after. The log is read-only, filterable and exportable.

Users and roles (Super Admin)

Register users and tick their roles (Admin, Stock Manager, Cashier, or a role the Super Admin built); edit or disable users; reset passwords and PINs; create and delete roles; see login history and active sessions; force a logout.

Settings

Shop details and logo, receipt header and footer in English or Sinhala, printers, document number formats (letters allowed), rounding, month start date, quotation validity, negative-stock rule (off), default shortcut keys, alert limits, backup schedule and folder, cloud link and language.

Extra features worth adding

- **Live today view online:** sales appear as they are billed.
- **Price history per product:** spot supplier price rises early.
- **Supplier comparison (Super Admin):** the sub-product model makes "who is cheapest for this item?" a single click.
- **Dead stock list with value:** frees money tied up in slow items.
- **Cash differences per cashier** over time.
- **Comparisons:** this month vs last month and vs the same month last year.
- **Daily summary** PDF sent to the owner at day end.
- **Monthly sales target** vs actual.

Keyboard shortcuts and auto-suggest

Every screen works with zero mouse: Enter moves forward, Esc goes back, arrows move in lists, and every button shows its key.

Your own keys: each user can remap any shortcut in their profile and save their own key map. The Super Admin sets the defaults below, and "Reset to default" brings them back. A key already used on that screen, or one the browser reserves, can't be assigned.

Defaults everywhere

Key	Action
Ctrl+K	Command palette: jump to any screen, product, supplier or bill
Alt+1 / Alt+2 / Alt+3	Billing / Stock / Admin
F1	Shortcut help for the current screen
Enter / Shift+Enter	Next field / previous field
Esc	Close, back or cancel
↑ ↓ and ← →	Move in lists, and between columns in grids
Space	Select a row (returns, reorder)
Ctrl+S	Save as draft
Ctrl+Enter	Post or confirm
Ctrl+F	Filter the current list
Ctrl+P	Print or export

Defaults in billing

Key	Action
F2	Item code field
F3	Customer name (optional, for quotations)
F4	Invoice or quotation
F6	Wholesale checkbox
F7 / F8	Hold bill / recall a held bill
F9	Bill history: P reprint, R return, C convert, V void
F10	Pay in cash, then Print receipt? Yes or No
Del	Remove the selected line
+ / −	Qty up or down on the selected line

Key	Action
L after the qty	Loose liters, for example 1.5L

F5, F11, F12 and Ctrl+W/T/N stay unused because browsers reserve them; the counter app also blocks them to prevent accidents.

Auto-suggest

Field	Searches by	Shows
Item code and Item name (billing)	Code, sub code, name or brand, typed in either field	Stock, price, unit
Item (purchases, transfers, adjustments, reorder, reports)	Code, sub code, name, brand	Stock, unit
Supplier	Code, name, phone	Balance
Sales rep	Name, only the chosen supplier's reps	Phone
Customer name (quotations)	Earlier names	Phone
Category, brand, unit, location	Name	"+ Add" when not found
Bill, quotation and GRN no	Any part, letters or digits	Customer or supplier, total
Description and note fields	Earlier entries	—

- Suggestions start from the first character for codes and the second for names.
- Matching forgives typos and word order: "thin 4" finds "Thinner 4L".
- Recent and frequent items come first. ↑↓ moves, Enter or Tab picks, Esc closes.
- Picking an item fills the active row with its code, description, price and default discount, ready for the qty.
- Results appear in under 100 ms on the shop network, using PostgreSQL trigram indexes.

Validation rules

Every value is checked twice: on the screen as it is typed, and again on the server before anything is saved.

General

- Required fields are marked, and Save or Post stays disabled until errors are fixed. Messages sit under the field in plain words.
- Money is 0 or more with 2 decimals. Quantities are more than 0, with up to 3 decimals only for L and kg; pieces and cans are whole numbers except in a loose sale.
- Codes are unique, ignoring capital letters and spaces, and may mix letters and digits. A similar name shows a warning, for example "A supplier with this name already exists".
- Document numbers (invoices, quotations, GRNs) accept letters and digits wherever they are typed or searched.
- Phone numbers: 10 digits starting with 0, or +94, stored in one format.
- Dates: no future dates except expiry and cheque dates, nothing posted into a closed month, and expiry after the purchase date. Expiry within 90 days shows a warning.

Purchases

- The same supplier with the same invoice no shows a warning before saving.
- Qty more than 0, Free 0 or more, Discount 0–100% (or not more than Value).
- Sales discount not above the purchase discount; sales price not above R-price and not below Cost.
- W-price between actual price and R-price.
- The net total must match the supplier's invoice total within Rs 1 to post.

Stock

- No location can go below zero.
- Transfers: from and to must differ, and qty can't exceed what the source holds.
- Mix paint: whole cans only, moved by Stock Managers or Admins. The price per liter can't go below the average cost, and a cashier changes it only with an Admin PIN.
- Adjustments need a reason; write-offs need Admin approval.

Billing

- Qty can't exceed the stock in the counter's location. This is checked again at payment, so two counters can't sell the last can.

- A line's discount can't exceed its batch's purchase discount, its price can't exceed the R-price, and no line sells below cost.
- Wholesale bills take each line's W-price and are marked Wholesale.
- The cash given must cover the net total.
- Converting a quotation re-checks stock and prices.
- Returns: qty up to sold minus already returned, and a refund no more than the cash paid for those units.

Admin

- The last Super Admin, a role still ticked for a user, and any supplier, product or value with history can't be deleted; they are deactivated instead.
- Passwords have at least 8 characters. PINs have 4–6 digits and are unique per user.
- A user's key map can't give one key two actions on the same screen.
- A month closes once; reopening needs the Super Admin and is logged.

Database safety net: check constraints (qty and prices not negative), unique indexes and foreign keys, so bad data can't get in even through a bug.

UI and UX guidelines

Calm, simple screens: one accent color, generous spacing, rounded corners, and only the controls each role needs.

- **Layout:** a left sidebar showing only the user's sections, a top bar with the page title and Ctrl+K search, and content in cards. Billing runs full screen with no sidebar.
- **Style:** light background, white cards, one accent color taken from the shop's brand, 12 px corners on cards and 8 px on inputs and buttons, soft borders, an 8 px spacing grid and plenty of white space.
- **Type:** Inter or Noto Sans (Noto Sans Sinhala if Sinhala is used), 15–16 px body text, 28–32 px totals, numbers right-aligned with even-width digits.
- **Formats:** Rs 1,250.00, dates as DD/MM/YYYY, and 12- or 24-hour time as a setting.
- **Color with meaning:** green for done and profit, amber for warnings such as low stock, red for errors such as below cost. Each always comes with an icon or words.
- **Key hints** on every button show the user's own keys, for example "Pay F10" and "Hold F7".
- **Forms:** labels above fields, one column on small screens and two on wide ones, errors under the field in plain words.

- **Tables:** sticky header, a clear focused row, filters on top, and the same export buttons everywhere.
- **Feedback:** small toasts for success, confirmation only for destructive actions, and no pop-up for anything that can be undone.
- **Empty states** say what to do next and show its shortcut.
- **Focus ring:** strong and always visible, because keyboard users must see where they are.
- **Dark mode** optional; contrast meets WCAG AA.
- **Speed targets:** any screen under 1 second, item search under 100 ms, and saving a bill under 300 ms on the shop network.

Data model overview

Stock and money are ledgers: every change is a row, and balances are worked out from those rows.

Area	Main tables
Access	users, roles, user_roles (the ticked checkboxes), role_permissions, user_shortcuts, sessions, audit_logs
Settings and lists	settings, number_formats, number_sequences, locations, categories, brands, units, reasons, cutoff_types
Suppliers	suppliers, supplier_reps, supplier_ledger, supplier_payments, cheques
Products	products (main, with box-to-piece conversion), sub_products, price_history
Stock	stock_balances (sub-product × location), stock_movements (every in and out, with cost and document link)
Purchasing	purchases, purchase_lines, purchase_returns, purchase_return_lines, reorder_lists, reorder_lines

Area	Main tables
Stock work	transfers, transfer_lines, mp_transfers, mp_links, adjustments, adjustment_lines, stock_counts, stock_count_lines
Sales	sales (invoice, quotation, held or void; wholesale flag, cash given, change, optional customer name), sale_lines (name, price and cost snapshot, loose details), sale_returns, sale_return_lines, reprints
Commission	commission_rules, commission_runs (once the rules are settled)
Cash	shifts (day ends), cash_movements
Month end	cutoff_rules, monthly_closings, monthly_closing_lines
System	sync_outbox, sync_state, backups, alerts, update_log

Every stock, bill and numbering row carries a location, so new shops or warehouses need no migration. Mix paint items are products of type MIX with one sub-product (the MP code) in liters at weighted-average cost; mp_links ties several sizes of one paint to one mix paint item.

Decisions

Every question is answered. The second round is first, then the 38 earlier decisions.

Decided in the second round, 25 Oct 2026

#	Topic	Decision
39	Commission	A rate per staff member on their net sales, set by the Super Admin
40	Cost visibility	Totals of cost, profit and margin: Super Admin only. Each product's cost, profit and margin: Stock Managers too

#	Topic	Decision
41	Supplier payments	Cash, cheque or bank transfer, with a cheque register
42	New can at the counter	A Stock Manager or Admin approves it with a PIN
43	Loose paint returns	Not allowed
44	Margin and profit	Use the Actual price, so free items count
45	Same-price batches	No pop-up; the oldest sells first
46	Future shops	Plan for both; a separate branch syncs through the cloud later
47	Wholesale discount	No extra discount on the W-price
48	Updates	A pop-up offers to restart now; if cancelled, the update installs at the next restart
49	Month start	The Super Admin sets the month start date

Decided on 25 Oct 2026

#	Topic	Decision
1	Credit	Cash only, no credit sales
2	W-price	Small Wholesale checkbox on the billing screen
3	Payment methods	Cash only
4	Tax	No VAT or tax invoices
5	Returns	No time limit; Cashiers, Admins and Super Admins process them
6	Petty cash	Small daily expenses from the drawer are allowed
7	New cost	Makes a new sub-product
8	Expiry	Each expiry date gets its own sub-product
9	Picking a batch	Oldest highlighted, cashier confirms; skipped when only one
10	More than one batch holds	Each batch sells at its own price

#	Topic	Decision
11	Margin	Sales price after discount − Actual price (free items included)
12	Discounts	Set at price setup, pre-filled at billing; the maximum is the purchase discount
13	Boxed items	Bought in boxes, sold singly
14	Item codes	Old codes kept on import; new items get 5 digits
15	Item cost	Transport and invoice discounts are not added
16	Purchase edits	Only while stock hasn't moved; otherwise an Admin reverses
17	#ref	A free-text note
18	Codes	MP- for all mixed paint
19	Mix paint	One per product; sizes can be linked to one mix paint item
20	Loose price	The mix paint price per liter
21	Mix paint price at the counter	Changing it up or down needs an Admin PIN
22	Loose quantity	Liters and part cans both work
23	Recipes	Not saved; only mix paint stock is kept
24	Moving cans into mix paint	Stock Managers and Admins only
25	Counters	1–2 to start; one PC is the server
26	Locations	One shop to start; the Super Admin adds more later
27	Printer	80 mm thermal with a cash drawer; A4 bills as PDF
28	Scanner	None
29	Language	English screens, Sinhala receipt option
30	Power and internet	Patchy internet; UPS recommended
31	Online	View only in version 1
32	Hosting	Small VPS or managed host

#	Topic	Decision
33	Cashier's sales view	Own counter's daily total only
34	Percentage cutoffs	On what's left after fixed cutoffs; each rule can choose its base
35	Month end	The Super Admin sets the month start date; the report is generated on that day
36	Cost and profit	Super Admin only; the Requirement Updates sheet overrides "Admins too" in the answers sheet
37	Data import	Excel templates plus a full stock count on go-live day
38	Old-system returns	Allowed, Admin only

The 12 requirement updates from the second sheet are built into the sections above.

Build phases and go-live

Build in three phases so the shop bills on the new system early, then add control and online features.

1. **Phase 1, go-live core**
   - Foundation: login, roles by checkbox, per-user shortcut keys, value manager, audit log, sync outbox (written from day one), backups, installer with automatic updates, and data import.
   - Stock: suppliers with activity log and payments, products and sub-products, purchases, stock view and mix-paint transfers.
   - Billing: invoices, quotations, multi-price pop-up, loose sales, cash payment with the print prompt, hold and recall, receipt printing, history, reprints, returns and day end.
2. **Phase 2, control and online**
   - Admin dashboard and charts, all reports with PDF and Excel, commission, monthly closing.
   - Reorder with supplier PDFs, adjustments and stock count, purchase returns, alerts.
   - Cloud sync and the view-only online admin panel.
3. **Phase 3, extras**

- More shops or warehouses with transfers between them, the daily summary to the owner, and online approvals.

Testing: unit tests for every formula (purchase line, discount cap, mix paint average cost, returns, monthly closing), end-to-end keyboard tests of the billing flow with Playwright, and a test restore of a real backup.

Go-live checklist

- [ ] Import suppliers, products, categories and brands from the current system, keeping the old codes
- [ ] Full stock count on go-live day, entered as opening stock with current cost and prices
- [ ] Enter opening balances for suppliers
- [ ] Register users, tick their roles and set PINs
- [ ] Test the receipt printer, the Sinhala receipt and the cash drawer
- [ ] Set the backup folder and USB copy, then do one test restore
- [ ] Train each cashier for an hour, with a printed card of their shortcut keys beside the keyboard
- [ ] Run alongside the old system for 3–7 days and compare day-end totals
- [ ] Switch over on the 1st of a month so monthly closing starts clean